Robert Leggieri

APPLIED AI | DATA SCIENCE | TECHNOLOGY LEADERSHIP

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SUMMARY

M.S. in Artificial Intelligence with project work in NLP, model evaluation, and AI integration. More than twenty years managing business systems and automating workflows. Seeking applied AI, NLP, and research software opportunities.

TECHNICAL SKILLS

Models / experimentation: Python, PyTorch, Hugging Face Transformers, scikit-learn, LoRA / QLoRA

Data / evaluation: NumPy, pandas, SQL / SQLite, grouped evaluation, calibration, pytest, Git

Software / integration: REST APIs, ETL, React, Node.js, Socket.IO, PowerShell

Graduate work: Generated nearly 1M grounded visual QA pairs; built MLP, transformer, and CNN waypoint models in simulation.

SELECTED AI AND SOFTWARE PROJECTS

NLP Robustness through Training-Data Analysis

Graduate NLP project, UT Austin | Paper

- Instrumented DeBERTa training on 392,702 MNLI examples across five epochs; implemented a Hugging Face callback, weighted dataset, Trainer override, and category-weighted cross-entropy.
- Improved HANS challenge-set accuracy from 77.49% to 78.59% (+1.10 percentage points); standard MNLI validation remained close to baseline (88.47% versus 88.91%).

PraxiCom: Faculty-Directed Voice Simulation

Academic prototype, UT Austin NURSING-AI Challenge, Spring 2025 | GitHub

- Individually built the PraxiCom prototype with React interfaces, a Node.js / Socket.IO relay, and Hume EVI voice integration, guided by nursing faculty.
- Added configurable YAML patient personas, a faculty-defined evaluation rubric, session monitoring, and local export of transcripts, chat, scores, and notes.

Text-Prosody Study and Evaluation Software

Graduate study and independent software project | Report | GitHub

- Designed the Alignment Discrepancy Score (ADS) and built a text/audio feature pipeline across 32,401 utterances from 189 interviews for PHQ-8 prediction.
- Compared ADS with component baselines: ROC AUC 0.609 versus 0.706 for text-only and 0.486 for audio-only. The results reinforced the need to test whether multimodal features add information beyond their individual components.
- Extended the study into grouped evaluation, calibration, and provenance software. A separate EmoBank text benchmark improved valence RMSE from 0.4276 to 0.3908.

BQE CORE Connector: Weekly Workload Planning

Professional project, GLW | GitHub

- Built a read-only BQE CORE connector feeding employee and project history into a weekly planning prototype for capacity reviews, overload checks, and waiting-work tracking.
- Connected BQE reference data with team-entered plans and day estimates; implemented OAuth PKCE, incremental SQLite sync, retries, deduplication, and validation.

PROFESSIONAL EXPERIENCE

Network Manager - Gutschick, Little & Weber, P.A.

Burtonsville, MD | July 2004 - Present

- Automate workflows with Python and PowerShell; integrate business systems and translate user requirements into practical technical solutions.
- Manage technology operations, security, vendors, and infrastructure planning for an engineering firm supporting more than 40 users.
- Organize and deliver software and AI training; create instructional videos and written guides, and provide demonstrations and one-on-one guidance.

EDUCATION

The University of Texas at Austin - M.S., Artificial Intelligence, GPA 4.0 | August 2026

University of Maryland Global Campus - B.S., Data Science, Summa Cum Laude | May 2024